HIMMAT SINGH

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PROFESSIONAL SUMMARY

B.Tech Computer Science student (2024–2028) at Poornima University with hands-on experience building production-ready Python tools and cloud-native applications. Developed ROBIN, a fully local AI assistant featuring voice I/O, semantic intent routing, and Windows automation — optimized to run on 6 GB RAM systems. Holds Google Cloud and Oracle Cloud certifications in Generative AI. Seeking an internship in Python development, backend engineering, or cloud engineering where I can contribute to real systems and grow alongside experienced engineers.

EDUCATION

Poornima University — B.Tech Computer Science *2024 – 2028 (Expected)*

Relevant Coursework: Data Structures & Algorithms, OOP, Database Management, Computer Networks, Cloud Computing

TECHNICAL SKILLS

Languages: Python, JavaScript, HTML/CSS, SQL, C++ (DSA)
Frameworks & Libraries: Flask, CustomTkinter, SpeechRecognition, Edge-TTS, psutil, pycaw, pywin32
AI & Automation: Ollama (local LLM inference), Prompt Engineering, Windows COM automation, Voice I/O pipeline
Cloud Platforms: Google Cloud Platform (Cloud Run, Vertex AI, Dataplex), Oracle Cloud Infrastructure (OCI)
Backend & Databases: REST API design, Flask web development, SQL schema design, JSON-based data storage
DevOps & Tools: Git, GitHub, VS Code, Linux (CLI), GitHub Actions, Multithreading & Async I/O

PROJECTS

ROBIN – AI Virtual Assistant *2024 – Present*

Python · Ollama · SpeechRecognition · Edge-TTS · CustomTkinter · pywin32 · psutil · pycaw · Git | github.com/Himmat1608/ROBIN-AI

- Architected a hybrid semantic routing layer with five deterministic intent routes (greetings, memory retrieval, memory extraction, system commands, AI reasoning), reducing average response latency from 2.8s to 45ms for command-class queries by bypassing LLM inference and calling system APIs directly.
- Reduced memory footprint to 180 MB by deploying Ollama with qwen2.5:3b for local LLM inference instead of cloud APIs, enabling fully offline operation and zero API costs on systems with as little as 6 GB RAM.
- Built a complete voice I/O pipeline integrating SpeechRecognition (configurable 80-threshold mic calibration), Edge-TTS, and PowerShell MediaPlayer for real-time speech synthesis — resolving Windows COM threading conflicts that broke pyttsx3 under multithreaded load.
- Implemented multithreading and async I/O to prevent UI freezes during 2.8s LLM inference cycles, maintaining responsiveness in the CustomTkinter GUI across 100+ system automation tasks.
- Engineered a JSON-based persistent memory system with continuity tracking, adaptive presence detection, meta-intent layer (CANCEL / SIMPLIFY / CONTINUE / CLARIFY), and user profile storage for personalized session behaviour.

SGPA & CGPA Calculator – Web Application *2024*

Python · Flask · HTML · CSS · JavaScript · SQL | github.com/Himmat1608/SGPA-CGPA-Calculator

- Developed a Flask web application implementing credit-weighted grade algorithms for accurate SGPA and CGPA calculation, adopted by students at Poornima University for academic performance tracking and course planning.
- Designed an efficient SQL schema with indexed queries, maintaining sub-200ms response times under normal classroom usage loads.
- Built a fully responsive interface in HTML, CSS, and JavaScript that renders correctly across desktop, tablet, and mobile — ensuring accessibility for all students regardless of device.

CERTIFICATIONS & CLOUD EXPERIENCE

Oracle Cloud Infrastructure (OCI) – Generative AI Professional | 2025 — *completed hands-on labs in LLM fundamentals, prompt engineering, RAG, and generative AI deployment.*

Google Cloud – Generative AI Leader Track | 2025 — *three skill badges earned through applied lab work:*
Cloud Run Skill Badge — deployed auto-scaling serverless applications on Google Cloud Run. **Dataplex Skill Badge —** explored data cataloging and metadata management on Google Cloud Dataplex. **Vertex AI Prompt Design Skill Badge —** designed and evaluated LLM prompt engineering strategies on Vertex AI.

ADDITIONAL INFORMATION

Languages: English (professional), Hindi (native) **Interests:** Systems programming, local AI tooling, open source development, DevOps automation **Open Source:** ROBIN is publicly available on GitHub under open-source licence.